

Practical 7

Aim: Practical: Theme, Custom Styles, Drawables.

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <RelativeLayout android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_margin="10dp"
        android:layout_weight="1">
        <ImageButton android:id="@+id/lbtn_t1_Up"
            android:layout_width="70dp"
            android:layout_height="70dp"
            android:layout_centerVertical="true"
            android:layout_alignParentLeft="true"
            android:onClick="IncreaseScore"
            style="@style/PlusButtons"/>
        <ImageButton android:id="@+id/lbtn_t1_DOWN"
            android:layout_width="70dp"
            android:layout_height="70dp"
            android:layout_alignParentRight="true"
            android:layout_centerVertical="true"
            android:onClick="DecreaseScore"
            style="@style/MinusButtons"/>
        <TextView android:id="@+id/tv_t1_score"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_centerVertical="true"
            android:layout_centerHorizontal="true"
            android:textSize="25dp"
            style="@style/ScoreText" android:text="0"/>
    <TextView
        android:id="@+id/tv_t1"
```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Team 1" android:textSize="50dp"
        style="@style/TeamText"
        android:layout_centerHorizontal="true"/>
</RelativeLayout>

<RelativeLayout android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_margin="10dp"
    android:layout_weight="1">
    <ImageButton android:id="@+id/lbtn_t2_UP"
        android:layout_width="70dp"
        android:layout_height="70dp"
        android:layout_centerVertical="true"
        android:layout_alignParentLeft="true"
        android:onClick="IncreaseScore"
        style="@style/PlusButtons" />
    <ImageButton android:id="@+id/lbtn_t2_DOWN"
        android:layout_width="70dp"
        android:layout_height="70dp"
        android:layout_centerVertical="true"
        android:layout_alignParentRight="true"
        android:onClick="DecreaseScore"
        style="@style/MinusButtons" />
    <TextView android:id="@+id/tv_t2_score"
        android:layout_width="wrap_content"

        android:layout_height="wrapandroid:layout_c
        enterVertical="true"
        android:layout_centerHorizontal="true"
        android:textSize="25dp"
        style="@style/ScoreText" android:text="0" />
    <TextView
        android:id="@+id/tv_t2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Team 2" android:textSize="50dp"
        style="@style/TeamText"
        android:layout_centerHorizontal="true"/>
</RelativeLayout>
</LinearLayout>

```

MainActivity.java:

```

package com.example.pr7;

import android.os.Bundle; import
android.view.View;
import android.widget.ImageButton; import
android.widget.TextView; import
androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity; import
androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat; public class

MainActivity extends AppCompatActivity {

    ImageButton t1_up,t2_up,t1_down, t2_down; TextView
    t1_score, t2_score;
    int t1=0,t2=0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        t1_score=findViewById(R.id.tv_t1_score); t2_score=findViewById(R.id.tv_t2_score);
    }
    public void IncreaseScore (View view) { int id =
        view.getId();
        if (id == R.id.lbtn_t1_Up) { t1 += 1;
            t1_score.setText(String.valueOf(t1));
        } else if (id == R.id.lbtn_t2_UP) { t2 += 1;
            t2_score.setText(String.valueOf(t2));
        }
    }
    public void DecreaseScore (View view) { int id =
        view.getId();
        if (id == R.id.lbtn_t1_DOWN) { t1 -= 1;
            t1_score.setText(String.valueOf(t1));
        } else if (id == R.id.lbtn_t2_DOWN) { t2 -= 1;
            t2_score.setText(String.valueOf(t2));
        }
    }
}

```

```
}
```

Themes.xml:

```
<resources xmlns:tools="http://schemas.android.com/tools">
  <style name="Base.Theme.Practical7" parent="Theme.Material3.DayNight.NoActionBar">
    <!-- Customize your base theme here -->
  </style>

  <style name="Theme.Pr7" parent="Base.Theme.Practical7">
    <!-- Additional customization for Theme.Pr7 -->
  </style>

  <style name="ScoreButtons" parent="Base.Theme.Practical7">
    <item name="android:background">@drawable/button_background</item>
  </style>
  <style name="PlusButtons" parent="ScoreButtons">
    <item name="android:src">@android:drawable/arrow_up_float</item>
  </style>
  <style name="MinusButtons" parent="ScoreButtons">
    <item name="android:src">@android:drawable/arrow_down_float</item>
  </style>
  <style name="ScoreText">
    <item name="android:textAppearance">@style/TextAppearance.AppCompat.Display
3</item>
  </style>
  <style name="TeamText">
    <item name="android:textAppearance">@style/TextAppearance.AppCompat.Display
1</item>
  </style>
</resources>
```

Button_background.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android" android:shape="oval">
  <stroke android:width="2dp" android:color="@color/design_default_color_primary"></stroke>
</shape>
```

Output:



Practical 8

Aim: Practical: Save user data in the database.

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_margin="10dp"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText android:id="@+id/nameEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Name" />

    <EditText android:id="@+id/ageEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/nameEditText"
        android:hint="Age" android:inputType="number" />

    <EditText android:id="@+id/emailEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/ageEditText"
        android:hint="Email" />

    <Button
        android:id="@+id/registerButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/emailEditText"
        android:text="Register" />

    <Button
        android:id="@+id/viewButton"
        android:layout_width="wrap_content"
```

```

        android:layout_height="wrap_content"
        android:layout_below="@id/registerButton"
        android:text="View Students" />

```

```
</RelativeLayout>
```

MainActivity.java:

```
package com.ssasit.p8;
```

```

import android.content.Intent;
import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText; import
android.widget.Toast;

```

```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity; import
androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

```

```

public class MainActivity extends AppCompatActivity { private
    StudentRepository studentRepository; @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main); studentRepository
        = new StudentRepository(this); studentRepository.open();

        final EditText nameEditText = findViewById(R.id.nameEditText);
        final EditText ageEditText = findViewById(R.id.ageEditText); final EditText
        emailEditText = findViewById(R.id.emailEditText); Button registerButton =
        findViewById(R.id.registerButton); Button viewButton =
        findViewById(R.id.viewButton);

        registerButton.setOnClickListener(new View.OnClickListener() { @Override
            public void onClick(View view) {
                String name = nameEditText.getText().toString();
                int age = Integer.parseInt(ageEditText.getText().toString()); String
                email = emailEditText.getText().toString();

                studentRepository.addStudent(name, age, email); Toast.makeText(MainActivity.this,
                "Student Registered!",

```

```

Toast.LENGTH_SHORT).show();
    }
    });

    viewButton.setOnClickListener(new View.OnClickListener() { @Override
        public void onClick(View view) {
            Intent intent = new Intent(MainActivity.this, ViewActivity.class); startActivity(intent);
        }
    });
}
}

```

ActivityView.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ViewActivity">

    <TextView android:id="@+id/studentListTextView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Student List" android:textSize="18sp" />

</RelativeLayout>

```

ViewActivity.java:

```

package com.ssasit.p8;

import android.database.Cursor;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity; public class

ViewActivity extends AppCompatActivity {

    private StudentRepository studentRepository;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState); setContentView(R.layout.activity_view);
    }
}

```

```

        studentRepository = new StudentRepository(this); studentRepository.open();

        TextView studentListTextView =
            findViewById(R.id.studentListTextView);

        Cursor cursor = studentRepository.getAllStudents(); StringBuilder builder =
            new StringBuilder();
        while (cursor.moveToNext()) {
            String name = cursor.getString(cursor.getColumnIndex("name")); int age =
                cursor.getInt(cursor.getColumnIndex("age"));
            String email = cursor.getString(cursor.getColumnIndex("email"));

            builder.append("Name: ").append(name)
                .append(", Age: ").append(age)
                .append(", Email: ").append(email)
                .append("\n");
        }
        cursor.close();

        studentListTextView.setText(builder.toString());
    }

    @Override
    protected void onDestroy() { super.onDestroy();
        studentRepository.close();
    }
}

```

Dbhelper.java:

```

package com.ssasit.p8;

// DatabaseHelper.java
import android.content.Context;
import android.database.sqlite.SQLiteDatabase; import
android.database.sqlite.SQLiteOpenHelper;

public class dbhelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "studentdb.db"; private static
    final int DATABASE_VERSION = 1;

    private static final String TABLE_CREATE = "CREATE TABLE
        students (" +

```

```
"id INTEGER PRIMARY KEY AUTOINCREMENT, " +
"name TEXT, " + "age
INTEGER, " +
"email TEXT);";
```

```
public dbHelper(Context context) {
    super(context, DATABASE_NAME, null, DATABASE_VERSION);
}

@Override
public void onCreate(SQLiteDatabase db) {
    db.execSQL(TABLE_CREATE);
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion)
{
    db.execSQL("DROP TABLE IF EXISTS students"); onCreate(db);
}
}
```

StudentRepository.java:

```
package com.ssasit.p8;

// StudentRepository.java
import android.content.ContentValues;
import android.content.Context; import
android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;

public class StudentRepository { private
    SQLiteDatabase database; private
    dbHelper dbHelper;

    public StudentRepository(Context context) { dbHelper =
        new dbHelper(context);
    }

    public void open() {
        database = dbHelper.getWritableDatabase();
    }

    public void close() {
        dbHelper.close();
    }
}
```

```

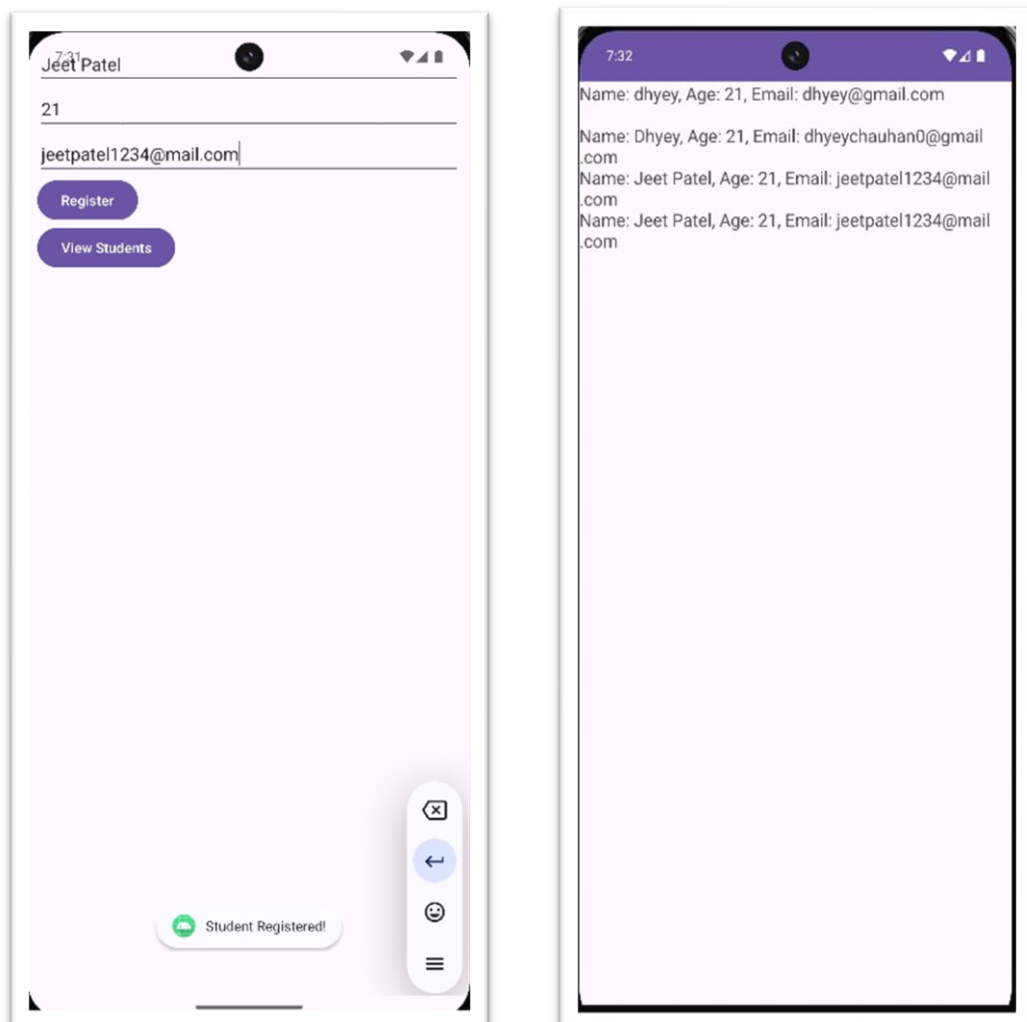
    }

    public void addStudent(String name, int age, String email) { ContentValues
        values = new ContentValues(); values.put("name", name);
        values.put("age", age);
        values.put("email", email); database.insert("students",
null, values);
    }

    public Cursor getAllStudents() {
        return database.query("students", null, null, null, null, null, null);
    }
}

```

Output:



Practical 9

Aim: Use an AsyncTask to access a remote database (make use of a simple PHP Web service).

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ListView android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java:

```
package com.ssasit.pr9;

import android.os.Bundle; import
android.util.Log;
import android.widget.ArrayAdapter; import
android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity; import
org.json.JSONArray;
import org.json.JSONObject;
public class MainActivity extends AppCompatActivity { private
    ListView listView;
    private ArrayAdapter<String> adapter; @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);
        listView = findViewById(R.id.listView);
        adapter = new ArrayAdapter<>(this, android.R.layout.simple_list_item_1);
        listView.setAdapter(adapter);
```

```

        String url = "http://192.168.1.6/MAD/index.php"; new
        FetchStudentsTask(this).execute(url);
    }
    public void updateUI(JSONArray jsonArray) { adapter.clear();
        Log.d("arraysize:",String.valueOf(jsonArray.length())); try {
            for (int i = 0; i < jsonArray.length(); i++) {
                JSONObject jsonObject = jsonArray.getJSONObject(i); String student =
                "ID: " + jsonObject.getInt("id") +
                ", Name: " + jsonObject.getString("name") + ", Age: "
                + jsonObject.getInt("age") +
                ", Email: " + jsonObject.getString("email"); adapter.add(student);
            }
        } catch (Exception e) { e.printStackTrace();
        }
    }
}

```

FetchStudentsTask.java:

```

package com.ssasit.pr9;

// FetchStudentsTask.java import
android.os.AsyncTask; import
android.util.Log; import
org.json.JSONArray;

import java.io.BufferedReader; import
java.io.InputStreamReader; import
java.net.HttpURLConnection;
import java.net.URL;

public class FetchStudentsTask extends AsyncTask<String, Void, String> { private static

    final String TAG = "FetchStudentsTask";
    private MainActivity activity;

    public FetchStudentsTask(MainActivity activity) {
        this.activity = activity;
    }

    @Override
    protected String doInBackground(String... params) { String
        response = "";

```

```

URLConnection urlConnection = null;

try {
    URL url = new URL(params[0]);
    urlConnection = (URLConnection) url.openConnection(); BufferedReader reader =
        new BufferedReader(new
InputStreamReader(urlConnection.getInputStream()));

    StringBuilder result = new StringBuilder(); String line;
    while ((line = reader.readLine()) != null) {
        result.append(line);
    }

    response = result.toString();
    Log.d(TAG, "Response: " + response); // Log response for debugging
} catch (Exception e) {
    Log.e(TAG, "Error fetching data", e);
} finally {
    if (urlConnection != null) { urlConnection.disconnect();
    }
}

return response;
}
@Override
protected void onPostExecute(String result) {
    super.onPostExecute(result);

    try {
        JSONArray jsonArray = new JSONArray(result); activity.updateUI(jsonArray);
    } catch (Exception e) {
        Log.e(TAG, "Error parsing JSON", e);
    }
}
}

```

Network_security_config.java:

```

<?xml version="1.0" encoding="utf-8"?>
<network-security-config>
    <domain-config cleartextTrafficPermitted="true">
        <domain includeSubdomains="true">192.168.1.6</domain>
    </domain-config>
</network-security-config>

```

Index.php:

```

<?php

$connect=mysqli_connect('localhost','root','','mad');

$query="SELECT * from student";

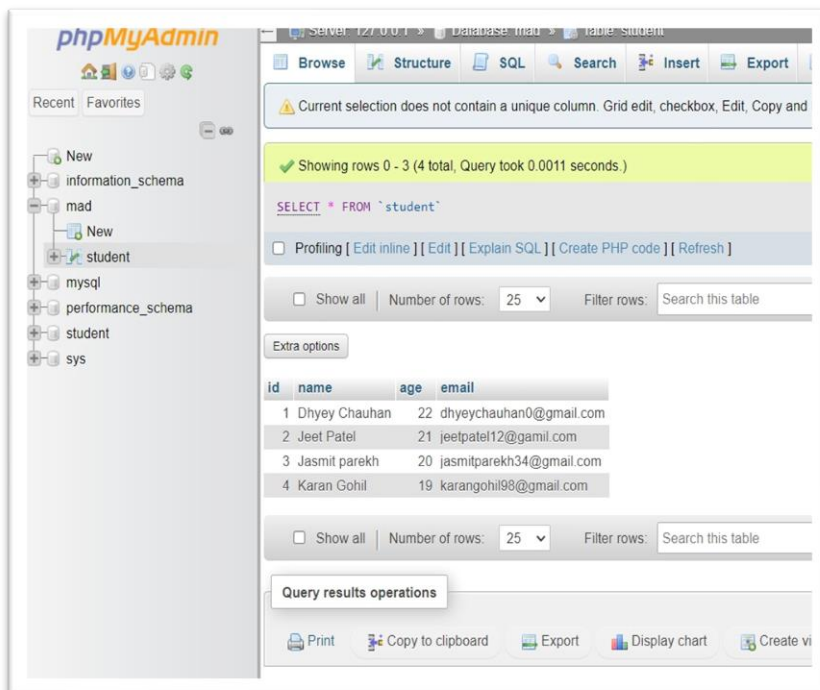
$result=mysqli_query($connect,$query);

while($i=mysqli_fetch_array($result))
{
    $data[] = array( 'id'
        => $i['id'],
        'name' => $i['name'],
        'age' => $i['age'],
        'email' => $i['email'],
    );
}

echo json_encode($data); mysqli_close($connect);

?>

```

Output:


The screenshot shows the phpMyAdmin interface with the 'student' table selected. The table contains 4 rows of data. The SQL query 'SELECT * FROM `student`' is entered in the SQL tab. The table structure is as follows:

id	name	age	email
1	Dhyey Chauhan	22	dhyeychauhan0@gmail.com
2	Jeet Patel	21	jeetpatel12@gamil.com
3	Jasmit parekh	20	jasmitparekh34@gmail.com
4	Karan Gohil	19	karangohil98@gmail.com



Practical 10

Aim: Use Retrofit to access remote database (make a use of simple PHP Web service).

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ListView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/listView" app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java:

```
package com.ssasit.pr10; import

android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity; import
androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.os.Bundle;

import android.util.Log;
import android.widget.AdapterView; import
android.widget.ListView; import
android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity; import
```

```

java.util.List;
import retrofit2.Call; import
retrofit2.Callback; import
retrofit2.Response; import
retrofit2.Retrofit;
import retrofit2.converter.gson.GsonConverterFactory; public

class MainActivity extends AppCompatActivity {

    private ListView listView;
    private ArrayAdapter<String> adapter; private
    ApiService apiService;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        // Initialize Retrofit
        //      Retrofit retrofit = RetrofitClient.getClient("https://192.168.3.25/MAD/");
        // Update with your server's URL
        Retrofit retrofit = new Retrofit.Builder()
            .baseUrl("https://raw.githubusercontent.com/devp22/json/main/")
            .addConverterFactory(GsonConverterFactory.create())
            .build();
        apiService = retrofit.create(ApiService.class);

        // Fetch students
        fetchStudents();
    }

    private void fetchStudents() {
        Call<List<Student>> call = apiService.getModels();
        call.enqueue(new Callback<List<Student>>() { @Override
            public void onResponse(Call<List<Student>> call, Response<List<Student>>
response) {
                if (response.isSuccessful() && response.body() != null) { List<Student>
                    students = response.body(); Log.d("Test", ""+students.size());
                    updateUI(students);
                } else {
                    Toast.makeText(MainActivity.this, "No data received",
Toast.LENGTH_SHORT).show();
                }
            }
        })
    }
}

```

```

    }

    @Override
    public void onFailure(Call<List<Student>> call, Throwable t) { Log.d("MainActivity",
        "Network Error: " + t.getMessage()); Toast.makeText(MainActivity.this, "Error fetching
        data",
        Toast.LENGTH_SHORT).show();

        // Optionally, print more detailed information
        t.printStackTrace();
    }
    });
}

private void updateUI(List<Student> students) {
    adapter = new ArrayAdapter<>(this, android.R.layout.simple_list_item_1);
    listView.setAdapter(adapter);

    for (Student student : students) {
        String studentInfo = "Name: " + student.getName() + ",
            Publisher: " + student.getPublisher() +
            ", NumberOfSongs: " + student.getNSongs(); adapter.add(studentInfo);
    }
}
}

```

RetrofitClient1.java:

```

package com.ssasit.pr10;

import retrofit2.Retrofit;
import retrofit2.converter.gson.GsonConverterFactory; public

class RetrofitClient1 {

    private static Retrofit retrofit = null;

    public static Retrofit getClient(String baseUrl) { if (retrofit
        == null) {
        retrofit = new Retrofit.Builder()
            .baseUrl(baseUrl)
            .addConverterFactory(GsonConverterFactory.create())
            .build();
        }
        return retrofit;
    }
}

```

```

    }
}

```

Student.java:

```

package com.ssasit.pr10;

public class Student {
//    private int id; private
    String Name;
    private String Publisher; private
    String NumberOfSongs;

    // Getters and setters

    public String getName() { return Name; }
    public void setName(String name) { this.Name = Name; }

    public String getPublisher() { return Publisher; }
    public void setPublisher(String age) { this.Publisher = Publisher; }

    public String getNSongs() { return NumberOfSongs; }
    public void setNSongs(String email) { this.NumberOfSongs = NumberOfSongs; }
}

```

ApiService.java:

```

package com.ssasit.pr10; import

android.util.Log; import

java.util.List;

import retrofit2.Call; import
retrofit2.http.GET;

public interface ApiService { @GET("newplaylist.json")
    Call<List<Student>> getModels();

}

```

Output:

Name: Meditation Music 2022, Publisher: Meditation Station, NumberOfSongs: 207
Name: Krishna Flute Meditation, Publisher: Nivya, NumberOfSongs: 25
Name: Morning Meditation, Publisher: thesnowman7, NumberOfSongs: 78
Name: Yoga and Meditation, Publisher: Anonymous, NumberOfSongs: 31
Name: Mindfulness and Meditation, Publisher: Billy-Joe, NumberOfSongs: 78
Name: 5 Minutes Meditation, Publisher: Betsy, NumberOfSongs: 42